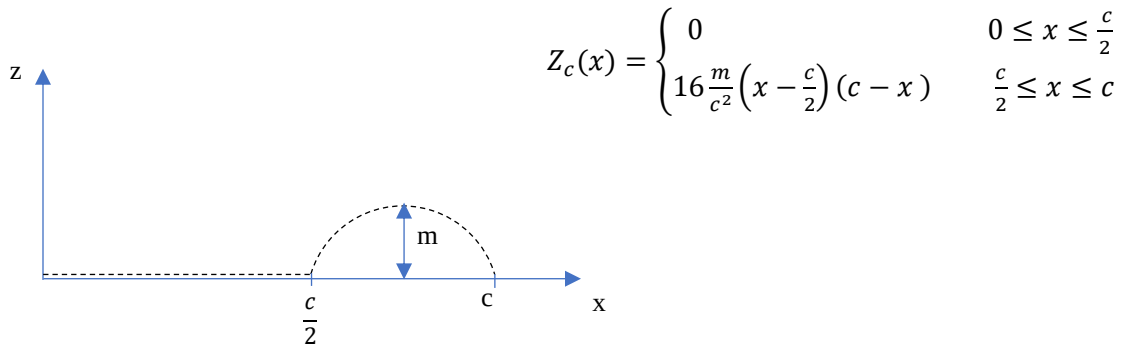


HW#5

Thin Airfoil Theory

- 1- The equation of the illustrated airfoil camber line is as follows:



The maximum camber value is equal to $m = 0.04c$.

- a. Using thin airfoil theory, show that the two coefficients A_1 and A_2 of this airfoil are obtained as follows:

$$A_1 = \frac{16}{\pi} \frac{m}{c} \left(\frac{\pi}{2} - 1\right) \quad A_2 = -\frac{32}{3\pi} \frac{m}{c}$$

- b. Calculate

The zero-lift angle of attack ($\alpha_{L=0}$),

The lift coefficient at 3-degree angle of attack.

The moment coefficient about the aerodynamic center.

- 2- The camber profile of an airfoil is given by:

$$\begin{cases} \frac{z}{c} = 0.2 \left(\frac{x}{c}\right) - 0.25 \left(\frac{x}{c}\right)^2 & 0 < x < 0.5 \\ \frac{z}{c} = \frac{3}{40} \left(1 - \frac{x}{c}\right) & 0.5 < x < 1 \end{cases}$$

Calculate

- a. The zero-lift angle of attack ($\alpha_{L=0}$),

- b. The lift force at an angle of attack of 10 degrees

- c. The moment coefficient around the leading edge at an angle of attack of 10 degrees

- e. The position of the center of pressure for the value of the coefficient calculated in part b

3- The equation of airfoil camber line is as follows:

$$z_c = ax(x - 1)(x - b)$$

where Z_c and x are dimensionless parameters using airfoil chord length (c) and a and b are dimensionless constants.

a. Find an equation for $\frac{dz_c}{dx}(\theta)$.

b. Find the constant value of b such that $C_{m_{ac}}=0$.

Hint: first write an equation for $C_{m_{ac}}$ in terms of A_n coefficient and then set it equal to zero.

c. If the value of $c_l=0.25$ and $\alpha=0$, find the value of a . Also calculate the value of $\bar{x}_{cp}$.

Hint:

$$\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$$

4- Consider an airfoil with a 2m chord in a flow with a velocity of 50 m/s at sea level. If the lift per unit span is 1353 N and the airfoil section has a curvature with the following characteristics, what is the angle of attack of this airfoil.

$$\frac{z}{c} = \begin{cases} 0.25 \left[0.8 \frac{x}{c} - \left(\frac{x}{c} \right)^2 \right] & \text{for } 0 \leq \frac{x}{c} \leq 0.4 \\ 0.111 \left[0.2 + 0.8 \frac{x}{c} - \left(\frac{x}{c} \right)^2 \right] & \text{for } 0.4 \leq \frac{x}{c} \leq 1 \end{cases}$$